Unit: Unit 0: Foundations of Algebra **Name:** ____________
Topic: Variables and Expressions / Order of Operations **Date:** ________

Challenge (Habanero)

Q1. A recipe for making cookies calls for $2 \cdot x + 5$ cups of flour, where x is the number of batches. If $x = 3$, how many cups of flour are needed?
(A) 11
(B) 15
(C) 16
(D) 20
(E) 21

Q2. A building has 5 floors, and each floor has $3 \cdot x - 2$ rooms, where x is the floor number. How many rooms are on the third floor?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q3. A chef needs to package 18 ingredients, with x ingredients per box. The equation for the number of boxes needed is $\frac{18}{x}$. If the chef wants to use 3 boxes, how many ingredients should each box contain?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q4. Simplify the expression: $2(x + 3) + 5(x - 2)$
(A) $7x + 1$
(B) $7x - 1$
(C) $7x - 6$
(D) $7x + 6$
(E) $7x - 4$

Q5. A carpenter is building a bookshelf with 5 shelves, and each shelf can hold $2x + 1$ books, where x is the shelf number. How many books can the second shelf hold?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q6. Evaluate the expression: $3 \cdot 2 + 12 \div 4$
(A) 7
(B) 8
(C) 9
(D) 10
(E) 11

Q7. A recipe for making bread calls for $x+2$ cups of flour, where x is the number of loaves. If the recipe makes 4 loaves, how many cups of flour are needed?
(A) 5
(B) 6
(C) 7
(D) 8
(E) 9

Q8. Simplify the expression: $(2x + 1) + (3x - 2)$
(A) $5x + 1$
(B) $5x - 1$
(C) $5x - 3$
(D) $5x + 3$
(E) $4x + 2$

Open Ended Questions (FRQ)

Q9. The length of a rectangular garden is given by the expression $2x + 5$, and the width is given by $x - 2$. Find the area of the garden when $x = 4$. Show your work and provide the final answer.

Q10. The cost of building a house is given by the expression $3000 + 500x + 200y$, where x is the number of bedrooms and y is the number of bathrooms. Find the cost of building a house with 3 bedrooms and 2 bathrooms. Show your work and provide the final answer.

Chef's Tips

- Tip 1: Make sure students understand the order of operations (PEMDAS)
- Tip 2: Encourage students to read the questions carefully and identify the variables and constants
- Tip 3: Remind students to show their work and provide explanations for free-response questions